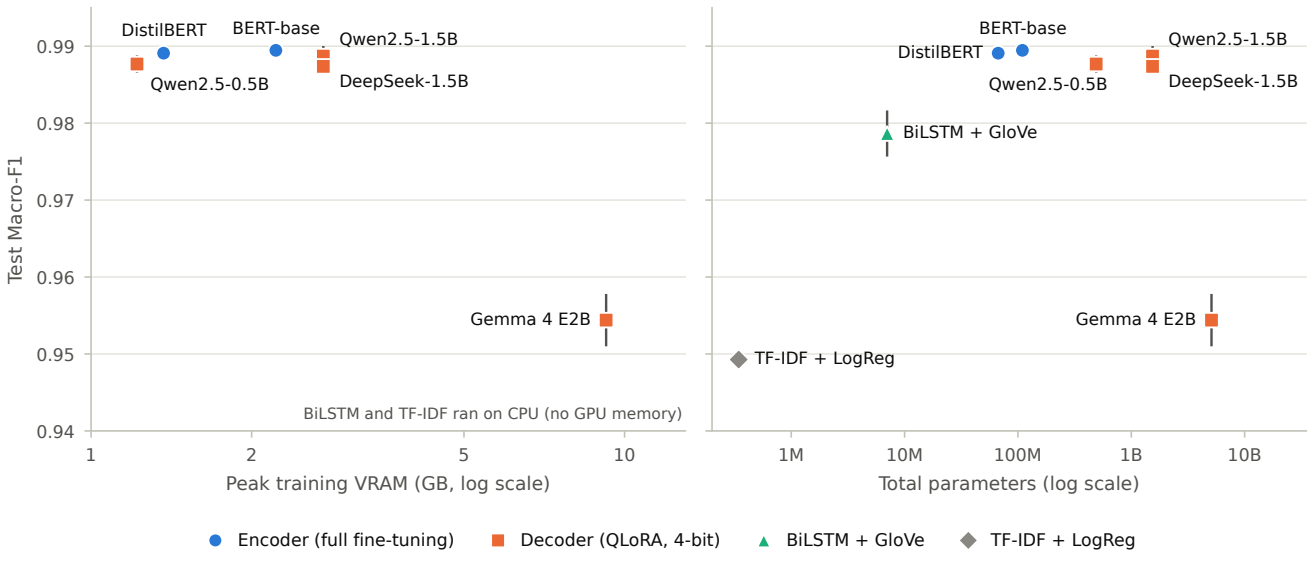
Test Macro-F1
0.99
0.98
0.97
0.96
0.95
0.94
DistilBERT
BERT-base
Qwen2.5-1.5B
Qwen2.5-0.5B
DeepSeek-1.5B
Gemma 4 E2B
BiLSTM and TF-IDF ran on CPU (no GPU memory)
1
2
5
10
Peak training VRAM (GB, log scale)
BERT-base
DistilBERT
Qwen2.5-1.5B
Qwen2.5-0.5B
DeepSeek-1.5B
BiLSTM + GloVe
Gemma 4 E2B
TF-IDF + LogReg
1M
10M
100M
1B
10B
Total parameters (log scale)
Encoder (full fine-tuning)
Decoder (QLoRA, 4-bit)
BiLSTM + GloVe
TF-IDF + LogReg